

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Cleaning Challenge

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO3: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.

Assessment Tool Weightage: 35%

Total Marks: 20

Code of Conduct

I **Yarra Pawan (192424294)** that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Y. Pawan**

Debugging

Criteria	Anomaly Detection (4 Marks)	Cleaning Methods (4 Marks)	Data Transformation (4 Marks)	Analytical Comparison (4 Marks)	Governance Discussion (4 Marks)
Marks					

Total Marks Awarded:

Questions:

Question 1: E-Commerce Customer Transactions Dataset

An online retail company has collected customer transaction data from multiple sales channels. The management wants to identify the top-performing customer segments and products. However, the dataset contains several quality issues.

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C101	Raj Kumar	Male	28	Electronics	2	15000 INR	10-01-2025	Chennai
C102	Priya S	Female	35	Fashion	3	1200 INR	11-01-2025	chennai
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
NULL	Sneha M	Female	30	Electronics	2	450 USD	05-02-2025	Trichy
C107	Deepak N	Male		Grocery	3	600 INR	07-02-2025	Tiruchirappalli
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem
C109	John D	Male	31	Electronics	4	250 EUR	15-02-2025	Chennai

Data Issues

- Missing Customer IDs
- Duplicate transactions
- Negative quantities
- Unit prices recorded in different currencies
- Age values such as 0, 150, and blanks
- Inconsistent city names (e.g., Chennai, chennai, CHENNAI)
- Incorrect purchase dates

Tasks

1. Identify and classify all data quality issues. (4 Marks)
2. Design a cleaning strategy for each issue. (4 Marks)
3. Perform data standardization and transformation. (4 Marks)
4. Analyze how cleaning affects total revenue calculations. (4 Marks)
5. Generate a report discussing business risks if the data were not cleaned (4 Marks)

Question 2: Hospital Patient Records Dataset

A hospital wants to analyze patient admissions and treatment effectiveness. Data were entered manually by different departments, leading to inconsistencies.

PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID
P1001	Ravi Kumar	45	M	10-01-2025	15-01-2025	Diabetes	25000	D101
P1002	Priya S	38	F	12-01-2025	11-01-2025	Hypertension	18000	D102
P1003	Arun V	125	M	13-01-2025	20-01-2025	diabetes	22000	D101
NULL	Kavya R	29	F	15-01-2025	17-01-2025	Diabtes	15000	D103
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1006	Sneha M	34	F	15-01-2025	18-01-2025	Hypertension	12000	D102
P1007	Deepak N		M	01-02-2025	05-02-2025	Asthma	10000	D104
P1008	Meena K	26	F	10-02-2025	15-02-2025	HYPERTENSION	13500	D102
P1009	John D	40	M	12-02-2025	18-02-2025	Diabetes	20000	D101

Data Issues

- Missing patient IDs
- Duplicate patient records
- Discharge dates before admission dates
- Null treatment costs
- Inconsistent diagnosis names
- Multiple formats of dates
- Unrealistic patient ages (>120)

Tasks

1. Detect all inconsistencies and anomalies. (4 Marks)
2. Develop rules for handling missing and invalid values. (4 Marks)
3. Create a process to standardize diagnosis names. (4 Marks)
4. Evaluate how cleaning changes average treatment cost and hospital stay duration. (4 Marks)
5. Recommend data governance measures to prevent future errors. (4 Marks)

Question 3: Banking Loan Approval Dataset

A bank uses customer data to determine loan eligibility. The dataset contains errors that may affect risk assessment and approval decisions.

CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
L001	750000	720	300000	Full Time	35	50000	Approved
L002	900K	680	400000	FT	42	100000	Approved
L003	1200000	NULL	600000	fulltime	28	200000	Approved
L004	150000	450	-200000	Part Time	30	50000	Rejected
L005	5000000	790	1000000	Self Employed	16	10000	Approved
L006	450000	350	250000	Unemployed	40	300000	Rejected
L007	75000000	800	500000	Full Time	45	100000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L010	NULL	710	300000	Full Time	37	80000	Approved

Data Issues

- Missing credit scores
- Income values recorded in different units
- Negative loan amounts
- Duplicate customer IDs
- Age values below 18
- Employment status entered differently (FT, Full Time, fulltime)
- Extreme outliers in income

Tasks

1. Identify critical issues that could impact loan decisions. (4 Marks)
2. Propose cleaning techniques for missing and inconsistent data. (4 Marks)
3. Detect and justify treatment of outliers. (4 Marks)
4. Analyze the effect of cleaning on loan approval rates. (4 Marks)
5. Discuss ethical implications of incorrect data cleaning decisions. (4 Marks)

Question 4: Smart City Traffic Monitoring Dataset

A smart city project collects traffic sensor data to optimize traffic signals. Sensor malfunctions and transmission errors have affected data quality.

SensorID	Timestamp	VehicleCount	AverageSpeed	TrafficSignalStatus	RoadName	WeatherCondition
S101	01-01-2025 08:00	120	45	Green	Anna Salai	Sunny
S102	01-01-2025 08:05	135	52	Red	anna salai	Sunny
S103	01-01-2025 08:10	-50	48	Green	ANNA SALAI	Cloudy
S104	01-01-2025 08:15	140	350	Yellow	GST Road	Rainy
S105	NULL	110	42	Green	GST ROAD	Sunny
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S107	01-01-2025 08:30	150	55	Green	Old Mahabalipuram Road	Sunny
S10A	01-01-2025 08:35	145	53	Green	OMR	Cloudy
S109	01-01-2025 08:40	138	50	Green	GST Road	Rainy

Data Issues

- Missing timestamps
- Duplicate sensor readings
- Vehicle counts recorded as negative numbers
- Average speed exceeding 300 km/h
- Missing weather information
- Inconsistent road naming conventions
- Sensor IDs with formatting errors

Tasks

1. Assess the quality of the traffic dataset. (4 Marks)
2. Recommend cleaning techniques for time-series data. (4 Marks)
3. Design rules to identify faulty sensor readings. (4 Marks)
4. Evaluate how cleaning affects congestion analysis. (4 Marks)
5. Suggest automated monitoring techniques for future data collection. (4 Marks)

Question 5: Global Sales Performance Dataset

A multinational company has merged sales data from five countries. Management wants to evaluate regional performance and forecast future sales. The merged dataset contains severe data quality problems.

Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating
Asia	India	Raj Kumar	Laptop	50000	INR	10-01-2025	4
Asia	India	Rajkumar	Laptop	55000	INR	10-01-2025	5
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
North America	USA	John D	Laptop	1500	USD	01-15-2025	6
Asia	India	Priya S	Mobile	NULL	INR	18-01-2025	3
Europe	France	Pierre L	Ordinateur Portable	1300	EUR	20-01-2025	4
Asia	India	Rahul P	Mobile	9999999	INR	25-01-2025	5
UK	UK	David T	Laptop	1100	GBP	28-01-2025	2
Asia	India	Priya S	Smartphone	35000	INR	30-01-2025	0

Data Issues

- Multiple currencies (USD, EUR, INR, GBP)
- Missing sales amounts
- Duplicate regional entries
- Customer ratings outside the scale (1–5)
- Different date formats across countries
- Product names in multiple languages
- Salespersons recorded with spelling variations
- Extreme sales outliers due to data-entry errors

Tasks

1. Develop a comprehensive data-cleaning framework for multinational datasets. (4 Marks)
2. Design methods to standardize currencies, products, and dates. (4 Marks)
3. Handle missing values and outliers using appropriate analytical techniques. (4 Marks)
4. Compare regional sales rankings before and after cleaning. (4 Marks)
5. Prepare an executive report explaining how data quality influences strategic business decisions. (4 Marks)

Signature of the Course Faculty

Question 1: E-Commerce Customer Transactions Dataset

Python Code

```
dataset = {
    "CustomerID": ["C101", "C102", "C103", "C104", "C105", "C105", None, "C107", "C108", "C109"],
    "CustomerName": ["Raj Kumar", "Priya S", "Arun V", "Kavya R", "Rahul P", "Rahul P", "Sneha M",
    "Deepak N", "Meena K", "John D"],
    "Gender": ["Male", "Female", "Male", "Female", "Male", "Male", "Female", "Male", "Female",
    "Male"],
    "Age": [28, 35, 150, 0, 42, 42, 30, 0, 27, 31],
    "ProductCategory": ["Electronics", "Fashion", "Electronics", "Grocery", "Fashion", "Fashion",
    "Electronics", "Grocery", "Fashion", "Electronics"],
    "Quantity": [2, 3, 1, -2, 5, 5, 2, 3, 1, 4],
    "UnitPrice": [15000, 1200, 300, 800, 2500, 2500, 450, 600, 1800, 250],
    "Currency": ["INR", "INR", "USD", "INR", "INR", "INR", "USD", "INR", "INR", "EUR"],
    "PurchaseDate": ["10-01-2025", "11-01-2025", "12-01-2025", "15-01-2025", "18-01-2025", "18-01-
    2025", "05-02-2025", "07-02-2025", "13-01-2025", "15-02-2025"],
    "City": ["Chennai", "chennai", "CHENNAI", "Coimbatore", "Madurai", "Madurai", "Trichy",
    "Tiruchirappalli", "Salem", "Chennai"]
}
data = pd.DataFrame(dataset)
print("Original Data")
print(data)
data = data.drop_duplicates()
data["CustomerID"] = data["CustomerID"].fillna("C106")
data.loc[data["Age"] < 18, "Age"] = data["Age"].median()
data.loc[:, "Age"] = data["Age"].astype(int)
Q1 = data["Age"].quantile(0.25)
Q3 = data["Age"].quantile(0.75)
IQR = Q3 - Q1
upper_bound = Q3 + (1.5*IQR)
data.loc[data["Age"] > upper_bound, "Age"] = data["Age"].median()
data.loc[data["Quantity"] < 0, "Quantity"] = data["Quantity"].abs()
data.loc[:, "City"] = data["City"].str.title()
city_mapping = {"Tiruchirappalli": "Trichy"}
data.loc[:, "City"] = data["City"].replace(city_mapping)
rates = { "INR": 1, "USD": 83, "EUR": 90}
data.loc[:, "UnitPrice"] = data["UnitPrice"] * data["Currency"].map(rates)
data["Total Price"] = data["UnitPrice"] * data["Quantity"]
data = data.drop(columns="Currency")
print("Total Revenue:", data["Total Price"].sum())
print("Revenue by Category:")
print(data.groupby("ProductCategory")["Total Price"].sum())
print("Revenue by City:")
```

```
print(data.groupby("City")["Total Price"].sum())
print("Final Cleaned Data")
data
```

Output:

Original Data

	CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	Currency	PurchaseDate	City
0	C101	Raj Kumar	Male	28	Electronics	2	15000	INR	10-01-2025	Chennai
1	C102	Priya S	Female	35	Fashion	3	1200	INR	11-01-2025	chennai
2	C103	Arun V	Male	150	Electronics	1	300	USD	12-01-2025	CHENNAI
3	C104	Kavya R	Female	0	Grocery	-2	800	INR	15-01-2025	Coimbatore
4	C105	Rahul P	Male	42	Fashion	5	2500	INR	18-01-2025	Madurai
5	C105	Rahul P	Male	42	Fashion	5	2500	INR	18-01-2025	Madurai
6	None	Sneha M	Female	30	Electronics	2	450	USD	05-02-2025	Trichy
7	C107	Deepak N	Male	0	Grocery	3	600	INR	07-02-2025	Tiruchirappalli
8	C108	Meena K	Female	27	Fashion	1	1800	INR	13-01-2025	Salem
9	C109	John D	Male	31	Electronics	4	250	EUR	15-02-2025	Chennai

Total Revenue: 240900
Revenue by Category:
ProductCategory
Electronics 219600
Fashion 17900
Grocery 3400
Name: Total Price, dtype: int64
Revenue by City:
City
Chennai 148500
Coimbatore 1600
Madurai 12500
Salem 1800
Trichy 76500
Name: Total Price, dtype: int64
Final Cleaned Data

	CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City	Total Price
0	C101	Raj Kumar	Male	28	Electronics	2	15000	10-01-2025	Chennai	30000
1	C102	Priya S	Female	35	Fashion	3	1200	11-01-2025	Chennai	3600
2	C103	Arun V	Male	30	Electronics	1	24900	12-01-2025	Chennai	24900
3	C104	Kavya R	Female	30	Grocery	2	800	15-01-2025	Coimbatore	1600
4	C105	Rahul P	Male	42	Fashion	5	2500	18-01-2025	Madurai	12500
5	C106	Sneha M	Female	30	Electronics	2	37350	05-02-2025	Trichy	74700
6	C107	Deepak N	Male	30	Grocery	3	600	07-02-2025	Trichy	1800
7	C108	Meena K	Female	27	Fashion	1	1800	13-01-2025	Salem	1800
8	C109	John D	Male	31	Electronics	4	22500	15-02-2025	Chennai	90000

Risks:

- 1. Missing Customer IDs can cause customer tracking problems.
- 2. Duplicate records can cause incorrect revenue.
- 3. Negative quantities can give wrong sales results.
- 4. Invalid ages can affect customer analysis.
- 5. Different currencies can affect revenue calculation.
- 6. Inconsistent city names can affect city-wise analysis.

Question 2: Hospital Patient Records Dataset

Python Code:

```
dataset = {
    "PatientID": ["P1001", "P1002", "P1003", None, "P1005", "P1005", "P1006", "P1007", "P1008",
    "P1009"],
    "Name": ["Ravi Kumar", "Priya S", "Arun V", "Kavya R", "Rahul P", "Rahul P", "Sneha M", "Deepak
    N", "Meena K", "John D"],
    "Age": [45, 38, 125, 29, 51, 51, 34, 50, 26, 40],
    "Gender": ["M", "F", "M", "F", "M", "M", "F", "M", "F", "M"],
    "AdmissionDate": ["10-01-2025", "12-01-2025", "13-01-2025", "15-01-2025",
    "18-01-2025", "18-01-2025", "15-01-2025", "01-02-2025",
    "10-02-2025", "12-02-2025"],
    "DischargeDate": ["15-01-2025", "11-01-2025", "20-01-2025", "17-01-2025",
    "25-01-2025", "25-01-2025", "18-01-2025", "05-02-2025",
    "15-02-2025", "18-02-2025"],
    "Diagnosis": ["Diabetes", "Hypertension", "diabetes", "Diabetes",
    "Asthma", "Asthma", "Hypertension", "Asthma",
    "HYPERTENSION", "Diabetes"],
    "TreatmentCost": [25000, 18000, 22000, 15000, None, None, 12000, 10000, 13500, 20000],
    "DoctorID": ["D101", "D102", "D101", "D103", "D104", "D104", "D102", "D104", "D102", "D101"]
}
data = pd.DataFrame(dataset)
print("Original Data")
print(data)
data = data.drop_duplicates()
data["PatientID"] = data["PatientID"].fillna("P1004")
data.loc[data["Age"] > 120, "Age"] = data["Age"].median()
data["AdmissionDate"] = pd.to_datetime(
    data["AdmissionDate"], dayfirst=True
)
data["DischargeDate"] = pd.to_datetime(
    data["DischargeDate"], dayfirst=True
)
data.loc[
    data["DischargeDate"] < data["AdmissionDate"],
    "DischargeDate"
] = data["AdmissionDate"]
data["TreatmentCost"] = data["TreatmentCost"].fillna(
    data["TreatmentCost"].median()
)
data["Diagnosis"] = data["Diagnosis"].str.title()
data["HospitalStay"] = (
    data["DischargeDate"] - data["AdmissionDate"]
)
```

```

).dt.days
print("Average Treatment Cost:")
print(data["TreatmentCost"].mean())
print("Average Hospital Stay:")
print(data["HospitalStay"].mean())
print("Treatment Cost by Diagnosis:")
print(data.groupby("Diagnosis")["TreatmentCost"].mean())
print("Hospital Stay by Diagnosis:")
print(data.groupby("Diagnosis")["HospitalStay"].mean())
print("Final Cleaned Data")
print(data)

```

Output:

Original Data									
	PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID
0	P1001	Ravi Kumar	45	M	10-01-2025	15-01-2025	Diabetes	25000.0	D101
1	P1002	Priya S	38	F	12-01-2025	11-01-2025	Hypertension	18000.0	D102
2	P1003	Arun V	125	M	13-01-2025	20-01-2025	diabetes	22000.0	D101
3	None	Kavya R	29	F	15-01-2025	17-01-2025	Diabetes	15000.0	D103
4	P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NaN	D104
5	P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NaN	D104
6	P1006	Sneha M	34	F	15-01-2025	18-01-2025	Hypertension	12000.0	D102
7	P1007	Deepak N	50	M	01-02-2025	05-02-2025	Asthma	10000.0	D104
8	P1008	Meena K	26	F	10-02-2025	15-02-2025	HYPERTENSION	13500.0	D102
9	P1009	John D	40	M	12-02-2025	18-02-2025	Diabetes	20000.0	D101

```
Average Treatment Cost:
16888.88888888889
Average Hospital Stay:
4.333333333333333
Treatment Cost by Diagnosis:
Diagnosis
Asthma      13250.0
Diabetes    20500.0
Hypertension 14500.0
Name: TreatmentCost, dtype: float64
Hospital Stay by Diagnosis:
Diagnosis
Asthma      5.500000
Diabetes    5.000000
Hypertension 2.666667
Name: HospitalStay, dtype: float64
Final Cleaned Data
```

	PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID	HospitalStay
0	P1001	Ravi Kumar	45	M	2025-01-10	2025-01-15	Diabetes	25000.0	D101	5
1	P1002	Priya S	38	F	2025-01-12	2025-01-12	Hypertension	18000.0	D102	0
2	P1003	Arun V	40	M	2025-01-13	2025-01-20	Diabetes	22000.0	D101	7
3	P1004	Kavya R	29	F	2025-01-15	2025-01-17	Diabetes	15000.0	D103	2
4	P1005	Rahul P	51	M	2025-01-18	2025-01-25	Asthma	16500.0	D104	7
6	P1006	Sneha M	34	F	2025-01-15	2025-01-18	Hypertension	12000.0	D102	3
7	P1007	Deepak N	50	M	2025-02-01	2025-02-05	Asthma	10000.0	D104	4
8	P1008	Meena K	26	F	2025-02-10	2025-02-15	Hypertension	13500.0	D102	5
9	P1009	John D	40	M	2025-02-12	2025-02-18	Diabetes	20000.0	D101	6

Recommendations:

- 1. Use unique Patient IDs.
- 2. Validate admission and discharge dates.
- 3. Handle missing treatment costs.
- 4. Standardize diagnosis names.
- 5. Validate patient ages.

Question 3: Banking Loan Approval Dataset

Python Code

```
import pandas as pd
dataset = {
    "CustomerID": ["L001", "L002", "L003", "L004", "L005", "L006", "L007", "L008", "L008", "L010"],
    "AnnualIncome": [750000, "900K", 1200000, 150000, 5000000, 450000, 7500000, 650000, 650000, None],
    "CreditScore": [720, 680, None, 450, 790, 350, 800, 690, 690, 710],
    "LoanAmount": [300000, 400000, 600000, -200000, 1000000, 250000, 500000, 350000, 350000, 300000],
    "EmploymentStatus": ["Full Time", "FT", "fulltime", "Part Time", "Self Employed", "Unemployed", "Full Time", "FT", "FT", "Full Time"],
    "Age": [35, 42, 28, 30, 16, 40, 45, 33, 33, 37],
    "ExistingDebt": [50000, 100000, 200000, 50000, 10000, 300000, 100000, 70000, 70000, 80000],
    "LoanStatus":
    ["Approved", "Approved", "Approved", "Rejected", "Approved", "Rejected", "Approved", "Approved", "Approved", "Approved"]
}
data = pd.DataFrame(dataset)
print("Original Data")
print(data)
data = data.drop_duplicates()
data["AnnualIncome"] = data["AnnualIncome"].astype(str).str.replace("K", "000")
data["AnnualIncome"] = pd.to_numeric(data["AnnualIncome"], errors="coerce")
data["AnnualIncome"] = data["AnnualIncome"].fillna(
    data["AnnualIncome"].median()
)
data["CreditScore"] = data["CreditScore"].fillna(
    data["CreditScore"].median()
)
data.loc[data["LoanAmount"] < 0, "LoanAmount"] = data["LoanAmount"].median()
data.loc[data["Age"] < 18, "Age"] = data["Age"].median()
data["EmploymentStatus"] = data["EmploymentStatus"].replace({
    "FT": "Full Time",
    "fulltime": "Full Time",
    "Part Time": "Part Time",
    "Self Employed": "Self Employed",
    "Unemployed": "Unemployed"
})
Q1 = data["AnnualIncome"].quantile(0.25)
Q3 = data["AnnualIncome"].quantile(0.75)
IQR = Q3 - Q1
upper_bound = Q3 + (1.5 * IQR)
data.loc[data["AnnualIncome"] > upper_bound, "AnnualIncome"] = data["AnnualIncome"].median()
```

```

approval_rate = (
    data["LoanStatus"].eq("Approved").mean() * 100
)
print("Loan Approval Rate:")
print(approval_rate)
print("Average Income:")
print(data["AnnualIncome"].mean())
print("Average Loan Amount:")
print(data["LoanAmount"].mean())
print("Final Cleaned Data")
print(data)

```

Output

Original Data								
	CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
0	L001	750000	720.0	300000	Full Time	35	50000	Approved
1	L002	900K	680.0	400000	FT	42	100000	Approved
2	L003	1200000	NaN	600000	fulltime	28	200000	Approved
3	L004	150000	450.0	-200000	Part Time	30	50000	Rejected
4	L005	5000000	790.0	1000000	Self Employed	16	10000	Approved
5	L006	450000	350.0	250000	Unemployed	40	300000	Rejected
6	L007	7500000	800.0	500000	Full Time	45	100000	Approved
7	L008	650000	690.0	350000	FT	33	70000	Approved
8	L008	650000	690.0	350000	FT	33	70000	Approved
9	L010	None	710.0	300000	Full Time	37	80000	Approved

Loan Approval Rate:

77.7777777777779

Average Income:

730555.5555555555

Average Loan Amount:

450000.0

Final Cleaned Data

	CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
0	L001	750000.0	720.0	300000	Full Time	35	50000	Approved
1	L002	900000.0	680.0	400000	Full Time	42	100000	Approved
2	L003	1200000.0	700.0	600000	Full Time	28	200000	Approved
3	L004	150000.0	450.0	350000	Part Time	30	50000	Rejected
4	L005	825000.0	790.0	1000000	Self Employed	35	10000	Approved
5	L006	450000.0	350.0	250000	Unemployed	40	300000	Rejected
6	L007	825000.0	800.0	500000	Full Time	45	100000	Approved
7	L008	650000.0	690.0	350000	Full Time	33	70000	Approved
9	L010	825000.0	710.0	300000	Full Time	37	80000	Approved

Ethical Considerations:

1. Incorrect data can lead to unfair loan decisions.
2. Invalid credit scores can affect customer eligibility.
3. Age errors can cause unfair approvals or rejections.
4. Data cleaning should not introduce bias.

Question 4: Smart City Traffic Monitoring Dataset

Python Code

```
dataset = {
    "SensorID": ["S101","S102","S103","S104","S105","S106","S106","S107","S10A","S109"],
    "Timestamp": ["01-01-2025 08:00","01-01-2025 08:05","01-01-2025 08:10",
        "01-01-2025 08:15",None,"01-01-2025 08:25",
        "01-01-2025 08:25","01-01-2025 08:30",
        "01-01-2025 08:35","01-01-2025 08:40"],
    "VehicleCount": [120,135,-50,140,110,130,130,150,145,138],
    "AverageSpeed": [45,52,48,350,42,46,46,55,53,50],
    "TrafficSignalStatus":
["Green","Red","Green","Yellow","Green","Red","Red","Green","Green","Green"],
    "RoadName": ["Anna Salai","anna salai","ANNA SALAI","GST Road","GST ROAD",
        "OMR","OMR","Old Mahabalipuram Road","OMR","GST Road"],
    "WeatherCondition":
["Sunny","Sunny","Cloudy","Rainy","Sunny",None,None,"Sunny","Cloudy","Rainy"]
}
data = pd.DataFrame(dataset)
print("Ooriginal Data")
print(data)
data = data.drop_duplicates()
data["Timestamp"] = pd.to_datetime(
    data["Timestamp"], dayfirst=True
)
data["Timestamp"] = data["Timestamp"].fillna(
    data["Timestamp"].median()
)
data.loc[data["VehicleCount"] < 0, "VehicleCount"] = data["VehicleCount"].median()
data.loc[data["AverageSpeed"] > 200, "AverageSpeed"] = data["AverageSpeed"].median()
data["RoadName"] = data["RoadName"].str.title()
data["RoadName"] = data["RoadName"].replace({
    "Anna Salai": "Anna Salai",
    "Gst Road": "GST Road",
    "Omr": "OMR",
    "Old Mahabalipuram Road": "OMR"
})
data["WeatherCondition"] = data["WeatherCondition"].fillna("Unknown")
data["SensorID"] = data["SensorID"].replace({
    "S10A": "S108"
})
print("Average Vehicle Count:")
print(data["VehicleCount"].mean())
print("Average Speed:")
```



```

print(data["AverageSpeed"].mean())
print("Vehicle Count by Road:")
print(data.groupby("RoadName")["VehicleCount"].mean())
print("Average Speed by Road:")
print(data.groupby("RoadName")["AverageSpeed"].mean())
print("Final Cleaned Data")
print(data)

```

Output

Oroiginal Data							
	SensorID	Timestamp	VehicleCount	AverageSpeed	TrafficSignalStatus	RoadName	WeatherCondition
0	S101	01-01-2025 08:00	120	45	Green	Anna Salai	Sunny
1	S102	01-01-2025 08:05	135	52	Red	anna salai	Sunny
2	S103	01-01-2025 08:10	-50	48	Green	ANNA SALAI	Cloudy
3	S104	01-01-2025 08:15	140	350	Yellow	GST Road	Rainy
4	S105	None	110	42	Green	GST ROAD	Sunny
5	S106	01-01-2025 08:25	130	46	Red	OMR	None
6	S106	01-01-2025 08:25	130	46	Red	OMR	None
7	S107	01-01-2025 08:30	150	55	Green	Old Mahabalipuram Road	Sunny
8	S10A	01-01-2025 08:35	145	53	Green	OMR	Cloudy
9	S109	01-01-2025 08:40	138	50	Green	GST Road	Rainy

Average Vehicle Count:
133.66666666666666
Average Speed:
49.0
Vehicle Count by Road:
RoadName
Anna Salai 130.000000
GST Road 129.333333
OMR 141.666667
Name: VehicleCount, dtype: float64
Average Speed by Road:
RoadName
Anna Salai 48.333333
GST Road 47.333333
OMR 51.333333
Name: AverageSpeed, dtype: float64
Final Cleaned Data

	SensorID	Timestamp	VehicleCount	AverageSpeed	TrafficSignalStatus	RoadName	WeatherCondition
0	S101	2025-01-01 08:00:00	120	45	Green	Anna Salai	Sunny
1	S102	2025-01-01 08:05:00	135	52	Red	Anna Salai	Sunny
2	S103	2025-01-01 08:10:00	135	48	Green	Anna Salai	Cloudy
3	S104	2025-01-01 08:15:00	140	50	Yellow	GST Road	Rainy
4	S105	2025-01-01 08:20:00	110	42	Green	GST Road	Sunny
5	S106	2025-01-01 08:25:00	130	46	Red	OMR	Unknown
7	S107	2025-01-01 08:30:00	150	55	Green	OMR	Sunny
8	S108	2025-01-01 08:35:00	145	53	Green	OMR	Cloudy
9	S109	2025-01-01 08:40:00	138	50	Green	GST Road	Rainy

- Recommendations:**
- 1. Validate sensor readings automatically.
 - 2. Check missing timestamps.
 - 3. Remove duplicate sensor records.
 - 4. Set valid limits for vehicle count and speed.
 - 5. Standardize road names.

Question 5: Global Sales Performance Dataset

Python Code

```
dataset = {
    "Region": ["Asia","Asia","Europe","Europe","North America","Asia","Europe","Asia","UK","Asia"],
    "Country": ["India","India","Germany","Germany","USA","India","France","India","UK","India"],
    "SalesPerson": ["Raj Kumar","Rajkumar","Anna M","Anna M","John D","Priya S","Pierre L","Rahul
P","David T","Priya S"],
    "Product": ["Laptop","Laptop","Laptop","Laptop","Laptop","Mobile","Ordinateur
Portable","Mobile","Laptop","Smartphone"],
    "SalesAmount": [50000,55000,1200,1200,1500,None,1300,9999999,1100,35000],
    "Currency": ["INR","INR","EUR","EUR","USD","INR","EUR","INR","GBP","INR"],
    "SalesDate": ["10-01-2025","10-01-2025","12-01-2025","12-01-2025","15-01-2025",
        "18-01-2025","20-01-2025","25-01-2025","28-01-2025","30-01-2025"],
    "CustomerRating": [4,5,4,4,6,3,4,5,2,0]
}
data = pd.DataFrame(dataset)
print("Original Data")
print(data)
data = data.drop_duplicates()
data["SalesAmount"] = data["SalesAmount"].fillna(
    data["SalesAmount"].median()
)
data["SalesDate"] = pd.to_datetime(
    data["SalesDate"], dayfirst=True
)
data["CustomerRating"] = data["CustomerRating"].clip(1,5)
data["SalesPerson"] = data["SalesPerson"].replace({
    "Rajkumar": "Raj Kumar"
})
data["Product"] = data["Product"].replace({
    "Ordinateur Portable": "Laptop",
    "Smartphone": "Mobile"
})
rates = {
    "INR": 1,
    "USD": 83,
    "EUR": 90,
    "GBP": 105
}
data["Sales_INR"] = (
    data["SalesAmount"] * data["Currency"].map(rates)
)
Q1 = data["Sales_INR"].quantile(0.25)
```

```

Q3 = data["Sales_INR"].quantile(0.75)
IQR = Q3 - Q1
upper_bound = Q3 + (1.5 * IQR)
data.loc[data["Sales_INR"] > upper_bound, "Sales_INR"] = data["Sales_INR"].median()
print("Total Sales in INR:",data["Sales_INR"].sum())
print("Sales by Region:",data["Sales_INR"].sum())
print("Sales by Country:")
print(data.groupby("Country")["Sales_INR"].sum())
print("Average Customer Rating:")
print(data["CustomerRating"].mean())
print("\nSales by Product:")
print(data.groupby("Product")["Sales_INR"].sum())
print("Final Cleaned Data")
print(data)

```

Output

Original Data								
	Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating
0	Asia	India	Raj Kumar	Laptop	50000.0	INR	10-01-2025	4
1	Asia	India	Rajkumar	Laptop	55000.0	INR	10-01-2025	5
2	Europe	Germany	Anna M	Laptop	1200.0	EUR	12-01-2025	4
3	Europe	Germany	Anna M	Laptop	1200.0	EUR	12-01-2025	4
4	North America	USA	John D	Laptop	1500.0	USD	15-01-2025	6
5	Asia	India	Priya S	Mobile	NaN	INR	18-01-2025	3
6	Europe	France	Pierre L	Ordinateur Portable	1300.0	EUR	20-01-2025	4
7	Asia	India	Rahul P	Mobile	9999999.0	INR	25-01-2025	5
8	UK	UK	David T	Laptop	1100.0	GBP	28-01-2025	2
9	Asia	India	Priya S	Smartphone	35000.0	INR	30-01-2025	0

```
Total Sales in INR: 731250.0

Sales by Region: 731250.0

Sales by Country:
Country
France      117000.0
Germany     108000.0
India       266250.0
UK          115500.0
USA         124500.0
Name: Sales_INR, dtype: float64

Average Customer Rating:
3.6666666666666665

Sales by Product:
Product
Laptop      570000.0
Mobile      161250.0
Name: Sales_INR, dtype: float64
```

Final Cleaned Data									
	Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating	Sales_INR
0	Asia	India	Raj Kumar	Laptop	50000.0	INR	2025-01-10	4	50000.0
1	Asia	India	Raj Kumar	Laptop	55000.0	INR	2025-01-10	5	55000.0
2	Europe	Germany	Anna M	Laptop	1200.0	EUR	2025-01-12	4	108000.0
4	North America	USA	John D	Laptop	1500.0	USD	2025-01-15	5	124500.0
5	Asia	India	Priya S	Mobile	18250.0	INR	2025-01-18	3	18250.0
6	Europe	France	Pierre L	Laptop	1300.0	EUR	2025-01-20	4	117000.0
7	Asia	India	Rahul P	Mobile	9999999.0	INR	2025-01-25	5	108000.0
8	UK	UK	David T	Laptop	1100.0	GBP	2025-01-28	2	115500.0
9	Asia	India	Priya S	Mobile	35000.0	INR	2025-01-30	1	35000.0

- Executive Report:**
- 1. Multiple currencies can affect global sales comparison.
 - 2. Missing sales values can reduce accuracy.
 - 3. Duplicate records can increase total sales incorrectly.
 - 4. Invalid ratings can affect customer satisfaction analysis.
 - 5. Inconsistent product names can affect product-wise analysis.
 - 6. Incorrect sales values can affect business decisions.